

Problem 15.3

If $d = 0.05$, calculate $a_{\overline{12}|}$.

Solution

Since $d = 0.05$, the discount factor is

$$v = 1 - d.$$

Thus,

$$v = 1 - 0.05 = 0.95.$$

The relationship between the effective interest rate i and the effective discount rate d is

$$i = \frac{d}{1 - d}.$$

Therefore,

$$i = \frac{0.05}{0.95} = \frac{1}{19}.$$

Using the annuity-immediate formula,

$$a_{\overline{12}|} = \frac{1 - v^{12}}{i}.$$

Substituting $v = 0.95$ and $i = 1/19$ gives

$$a_{\overline{12}|} = \frac{1 - (0.95)^{12}}{1/19}.$$

Finally,

$$a_{\overline{12}|} \approx 8.7331583.$$

Hence,

$$\boxed{a_{\overline{12}|} \approx 8.7331583}.$$